Test1 Review, MAT 1575 Professor Chiu

1. True or False questions:

a) ____ $\int_a^b [cf(x) + dg(x)]dx = c\int_a^b f(x)dx + d\int_a^b g(x)dx$ where a, b, c, d are constants and f, g are integrable.

b) _____ The Riemann Sum is called Upper Sum if $f(x_i^*) \geq f(x)$ which is the maximum value of $f(x)$ for $x \in [x_{i-1}, x_i]$.

c) _____ $\frac{d}{dr}\int_0^r \sqrt{x^2+4}\,dx = \sqrt{r^2+4}$

d) _____ $\int_3^6 \sqrt{9-(x-3)^2}\ dx = \frac{9}{4}\pi$

e) _____ $\int_a^b f(x)^2 dx = \int_a^b f(x)dx \int_a^b f(x)dx$

2. Find the area between the curve of the function $f(x) = 2x - 1$ and the x-axis over the interval $[-1, 3]$.

3. Use the definition of the definite integral to evaluate $\int_{-1}^{3}(2x-1)dx$.

4. Evaluate $\int e^x \sin(x)\,dx$.

5. Evaluate $\int \tan^{-1}\theta\,d\theta$.

6. Evaluate $\int_{-\pi}^{\pi} \theta \sin\theta\,d\theta$

7. Find the antiderivative of the function $(\tan\theta)\ln(\cos\theta)$.

8. Find the antiderivative of the function $(\cos\theta)\ln(\sin\theta)$.

9. Evaluate $\int_0^1 x^2(x^3+1)^3\ dx$

10. Evaluate $\int_0^1 \frac{3x^2}{\sqrt[3]{x^3+1}}\,dx$

11. Evaluate $\int x^2\ln(x)\ dx$

12. Evaluate $\int x^2 e^{-x}\ dx$

13. Evaluate $\int x\cos(3x)\ dx$

14. Evaluate $\int_1^4 \frac{2-\sqrt{t}}{t^2}dt$